$$\langle x\rangle_{T_s} = \frac{1}{T_s}\int_t^{t+T_s} x(\tau)d\tau \tag{5.3}$$

where $\langle x\rangle_{T_s}$ denotes the state-average value.

By using this approach, the switching function is the same with duty cycle, and state equations of inductor current and the capacitor voltage of Eq. (5.2) become,

$$\begin{cases} \begin{bmatrix} \langle u_{ga}\rangle_{T_s} \\ \langle u_{gb}\rangle_{T_s} \\ \langle u_{gc}\rangle_{T_s} \end{bmatrix} = L_g \frac{d}{dt}\begin{bmatrix} \langle i_{ga}\rangle_{T_s} \\ \langle i_{gb}\rangle_{T_s} \\ \langle i_{gc}\rangle_{T_s} \end{bmatrix} + \begin{bmatrix} d_a - \frac{d_a+d_b+d_c}{3} \\ d_b - \frac{d_a+d_b+d_c}{3} \\ d_c - \frac{d_a+d_b+d_c}{3} \end{bmatrix} \langle v_{dc}\rangle_{T_s} \\ C\frac{d}{dt}\langle v_{dc}\rangle_{T_s} = [d_a \quad d_b \quad d_c]\begin{bmatrix} \langle i_{ga}\rangle_{T_s} \\ \langle i_{gb}\rangle_{T_s} \\ \langle i_{gc}\rangle_{T_s} \end{bmatrix} - \langle i_{load}\rangle_{T_s} \end{cases} \tag{5.4}$$

where d denotes the duty cycle. In this way, the circuit becomes continuous, while the switching ripple is removed compared with the real system.

5.1.3 Coordinate Transformation and Space Vector

With Clarke transformation, the three-phase static abc reference frame can be converted into the two-phase static $\alpha\beta$ reference frame, and vice versa,

$$C_{32} = \frac{2}{3}\begin{bmatrix} 1 & -\frac{1}{2} & -\frac{1}{2} \\ 0 & \frac{\sqrt{3}}{2} & -\frac{\sqrt{3}}{2} \end{bmatrix} \quad C_{23} = \begin{bmatrix} 1 & 0 \\ -\frac{1}{2} & \frac{\sqrt{3}}{2} \\ -\frac{1}{2} & -\frac{\sqrt{3}}{2} \end{bmatrix} \tag{5.5}$$

As shown in Fig. 5.2A, Eq. (5.5) is obtained by the projection of abc coordinate to $\alpha\beta$ coordinate. Meanwhile, a gain of 2/3 is used in order to keep the same amplitude.

Park transformation changes the static $\alpha\beta$ reference frame into the rotating dq reference frame, and vice versa,

$$C_{2s/2r} = \begin{bmatrix} \cos\omega t & \sin\omega t \\ -\sin\omega t & \cos\omega t \end{bmatrix} \quad C_{2r/2s} = \begin{bmatrix} \cos\omega t & -\sin\omega t \\ \sin\omega t & \cos\omega t \end{bmatrix} \tag{5.6}$$

where θ is the angle between the d-axis and α-axis, and ω is the angular speed of the grid. As shown in Fig. 5.2B, Eq. (5.6) is obtained by the projection of $\alpha\beta$ coordinate to dq coordinate